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A vision–language foundation model for the generation of realistic chest X-ray images

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The paucity of high-quality medical imaging datasets could be mitigated by machine learning models that generate compositionally diverse images that faithfully represent medical concepts and pathologies. However, large vision–language models are trained on natural images, and the diversity distribution of the generated images substantially differs from that of medical images. Moreover, medical language involves specific and semantically rich vocabulary. Here we describe a domain-adaptation strategy for large vision–language models that overcomes distributional shifts. Specifically, by leveraging publicly available datasets of chest X-ray images and the corresponding radiology reports, we adapted a latent diffusion model pre-trained on pairs of natural images and text descriptors to generate diverse and visually plausible synthetic chest X-ray images (as confirmed by board-certified radiologists) whose appearance can be controlled with free-form medical text prompts. The domain-adaptation strategy for the text-conditioned synthesis of medical images can be used to augment training datasets and is a viable alternative to the sharing of real medical images for model training and fine-tuning.

Latent diffusion models (LDMs) are a type of denoising diffusion probabilistic model, allowing for high-quality and diverse image generation[1]. When coupled with a conditioning mechanism (for instance, by cross-attention), these models enable fine-grained control over the image generation process during inference[1–3]. Stable Diffusion (SD) is an LDM that was pre-trained on a large multi-modal dataset (LAION-5B) of billions of natural image–text pairs[4]. Owing to its adaptability to various downstream tasks[1], SD can be considered a foundation model[5]. Since LDMs conduct their denoising process in a relatively low-dimensional latent space rather than a high-dimensional pixel space[1], they can operate on modest hardware resources, making them accessible to a wide range of end users.

In domains where access to high-quality training data is limited, the generative capabilities of LDMs can be particularly advantageous. Medical imaging is one such domain where the availability of well-curated datasets is often constrained by various challenges, such as the need for annotations by trained medical experts or compartmentalization within hospitals due to privacy concerns. Generative models such as LDMs capable of creating synthetic datasets with high fidelity could augment traditional supervised machine learning pipelines and potentially overcome some of these limitations.

Medical imaging studies are usually accompanied by a text report composed by radiologists or other imaging specialists, describing and interpreting relevant information contained within the images in detail. Previously, radiology reports have been used to generate labels for downstream classification tasks[6]. However, a report contains far more granular information than simple class labels and can be

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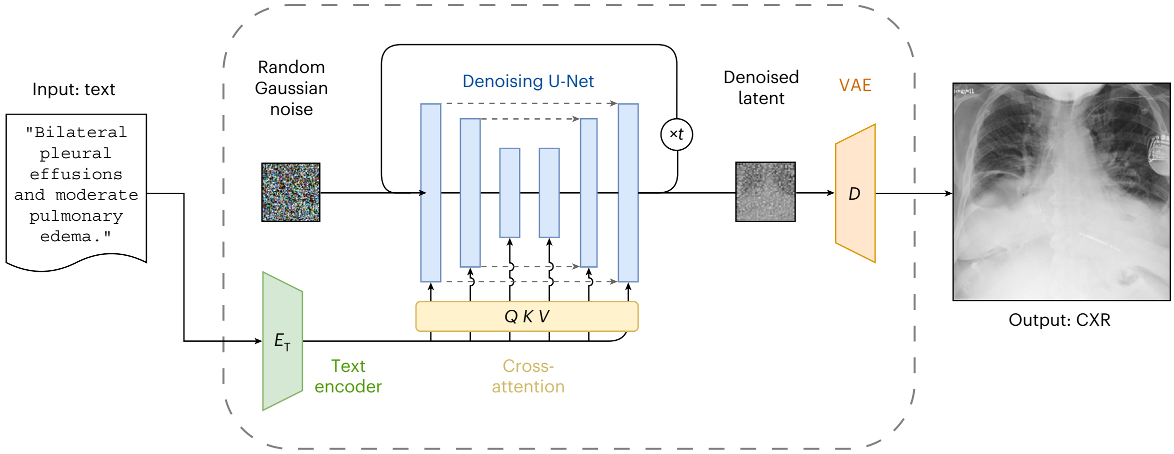

Fig. 1 | Text-to-image synthesis of CXR images using RoentGen, a medical domain-adapted LDM based on the SD pipeline. A conditional denoising U-Net iteratively denoises a latent vector sampled from a Gaussian distribution over t timesteps. The process is conditioned via cross-attention (*QKV* for query, key, value of the attention process) through embeddings created from short medical free-text inputs processed by a text encoder E_T. The decoder D of the VAE maps the denoised latent vector to pixel space, resulting in a high-fidelity CXR image showing imaging features corresponding to the initial text prompt.

used for more advanced approaches such as self-supervised learning (SSL)[7,8]. With medical image–text pairs, leveraging the vision–language entanglement of text-conditional LDMs may allow generating synthetic medical imaging data by prompting with relevant medical keywords or concepts of interest.

Given the opportunities that LDMs provide in medical imaging, in this study we explored adapting a large, general-domain pre-trained vision–language LDM (SD) to the medical domain and evaluated its representational quality. Benefitting from the extensive image–text pre-training underlying the components of the SD pipeline, we explored its use for generating chest X-rays (CXRs), one of the most commonly used radiological imaging modalities, conditioned on short radiology-specific text prompts.

Specifically, we systematically explored the domain adaptation of an LDM for the language-conditioned generation of medical images beyond the few-shot or zero-shot setting. We evaluated the representational capacity of the SD pipeline, exploring different strategies for improving this general-domain pre-trained foundational model for representing medical concepts specific to CXRs. In addition, we introduce a comprehensive evaluation framework to assess medical domain-adapted text-to-image models, exploring facets such as image quality, text–image alignment, representational fidelity for deep learning and factual correctness. By using this framework, we present evidence that fine-tuning both the U-Net and the CLIP (Contrastive Language-Image Pre-training[9]) text encoder yields the highest image fidelity and conceptual correctness for adapting the general-domain pre-trained LDM; that the synthetic images convey the concepts of the text prompts as confirmed by board-certified radiologists, pre-trained image classifiers and radiology report generation (RRG) metrics; and that text-conditioned synthetic CXRs can be used as a data augmentation resource. Training classification models on a blend of real and synthetic data led to slightly improved classification performance on in- and out-of-distribution data. This effect was comparable to self-supervised pre-training on the same amount of real CXRs. Also, we publicly released our best-performing model, dubbed 'RoentGen' (a portmanteau of 'Roentgen', as an homage to the pioneer of radiology, Wilhelm Conrad Röntgen, and 'Generator'), to provide the research community with a vision–language model for text-conditioned CXR generation.

Results

We fine-tuned the LDM SD using a large dataset of real CXRs and corresponding text reports obtained from MIMIC-CXR[10] with the goal to develop a generative model capable of synthesizing CXRs conditioned on text input. To this end, we explored various approaches for the SD pipeline, including fine-tuning different pipeline components, training on different radiological views (that is, orientations or angles from which a radiograph was acquired, such as 'frontal' or 'lateral'), hyperparameters such as learning rate and training steps, and using domain-specific text encoders instead of the default one (CLIP ViT-L/14).

The best-performing fine-tuned model, referred to as RoentGen, obtained the ability for text-conditioned synthesis of CXRs closely resembling their real counterparts (Fig. 1). To evaluate the quality of the generated CXRs, we employed a comprehensive evaluation framework comprising several steps: (1) board-certified radiologists conducted visual assessments of the generated CXRs; (2) image quality was assessed quantitatively using established metrics for fidelity and diversity; (3) convolutional neural networks (CNNs) pre-trained on real CXRs were used for multi-label classification performance assessment; (4) the factual correctness of radiological concepts in the synthetic CXRs was assessed using pre-trained multi-modal models for the tasks of RRG, image–image and image–text retrieval. Additionally, we assessed the value of using synthetic CXRs as a training resource for data augmentation and SSL and evaluated the performance of the resulting models on an in-distribution and two out-of-distribution test sets.

Radiologist assessment

Two board-certified radiologists inspected the generated CXRs. Using simple text prompts based on the 14 CheXpert labels (for instance, 'pneumothorax' or 'pleural effusion'), we find that a single domain-adapted SD model can create synthetic CXRs that clearly show the corresponding radiological findings (Fig. 2). Adding free-form textual modifiers for features such as abnormality extent or laterality (for example, 'big', 'moderate' and 'right-sided') leads to radiologically plausible correlates in the generated CXRs (Fig. 3). Multiple abnormalities can be combined in the same CXR, resulting in fine-grained compositionality (Fig. 4).

To assess the alignment between the text prompts and the generated visual radiological concepts, a reader study was conducted for synthetic CXRs generated by prompting the fine-tuned SD model with radiology report impression sections derived from MIMIC-CXR. The radiologists rated the alignment on a scale from −2 (indicating a complete mismatch between the text and the image) to 2 (representing a perfect text–image alignment, as discussed in Methods). Average ratings of 0.41 ± 1.41 and 0.29 ± 1.36 (mean ± standard deviation) indicated that the synthetic CXRs generally displayed most, but not always all,

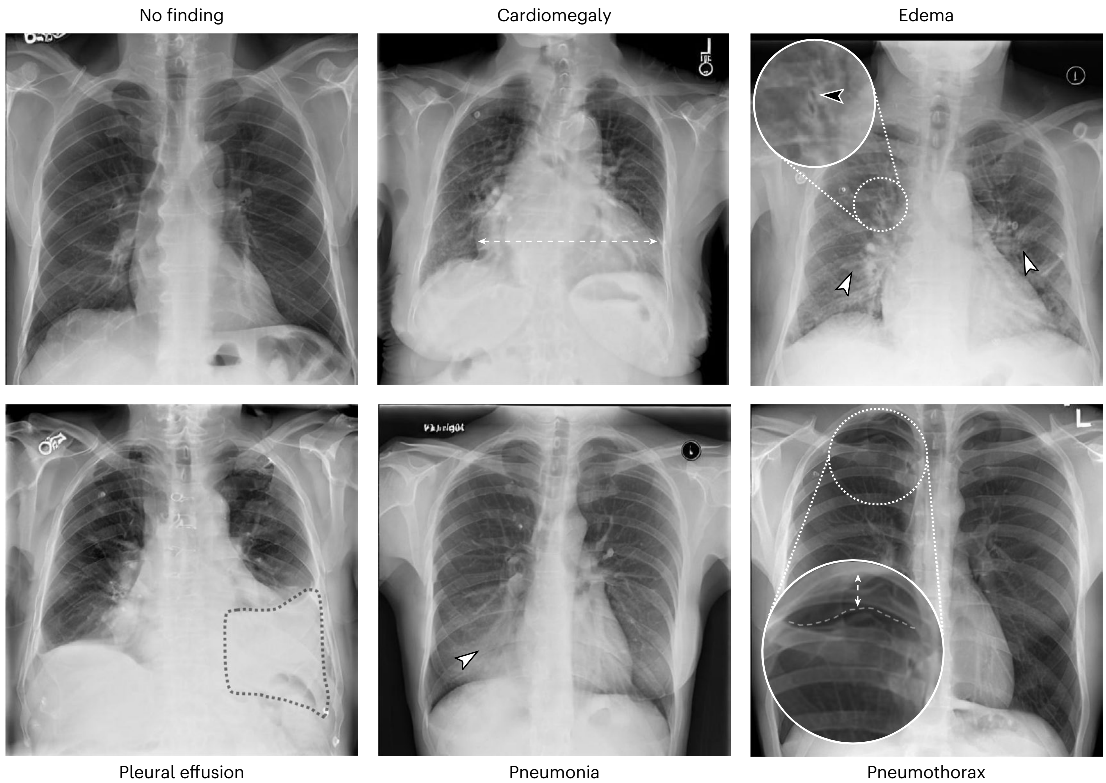

Fig. 2 | Text-conditional synthesis of CXRs. Samples were created by prompting a fine-tuned model (60k training steps; learning rate 5×10^{-5}; PA view) for typical CXR abnormalities. The generated CXRs feature high levels of detail: when prompted for 'edema' (top right), perihilar haziness (white arrowheads) and peribronchial cuffing (black arrowhead), both features seen in pulmonary oedema, can be observed. For 'pneumothorax' (bottom row, right image), a fine line representing the visceral pleural lining of the partially collapsed lung can be delineated (dashed line). The dotted regions were added for visualization.

of the findings mentioned in the impression section. This observation was particularly noticeable in cases with longer impression sections mentioning multiple findings. Nevertheless, these results underscore the model's ability to largely adhere to the conditional prompt.

Generative fidelity and diversity

Ideally, synthetic CXRs should closely resemble the distribution they are modelled after (fidelity) while at the same time covering as much of the variety of the underlying real images as possible (diversity).

Image fidelity was assessed quantitatively by computing the Fréchet inception distance (FID) between real and synthetic CXRs, using latent features extracted by a pre-trained CXR classification model (XRV[11]). Diversity was assessed by computing the pairwise multi-scale structural similarity index metric (MS-SSIM) (Table 1 and Extended Data Fig. 1). A smaller FID indicates that the generated images are more similar to the original images, whereas a lower MS-SSIM suggests higher diversity among the outputs. FID and MS-SSIM scores should be jointly considered when comparing synthetic image quality, as it is possible to create highly diverse yet unrealistic images.

All fine-tuning approaches, including runs with as little as 1k training steps, improved CXR fidelity over the original SD model (FID_{XRV} of 47.7) and the few-shot baseline (FID_{XRV} of 19.5). Increasing the number of training steps to 12.5k led to higher FID but lower MS-SSIM scores, indicating a tendency to produce more diverse but less realistic outputs during that stage. The highest fidelity was reached by training for 60k steps with a small learning rate (FID_{XRV} of 3.6). Over 1k training steps, randomly initializing the U-Net and jointly training it with the text encoder led to a 50% deterioration compared with the continuous fine-tuning equivalent, while still outperforming the baselines (FID_{XRV} of 9.0). Switching the default general-domain CLIP text encoder with a domain-specific text encoder like RadBERT[12] or SapBERT[13] slightly improved performance when the text encoder was kept frozen, and the U-Net was initialized from scratch. After 60k training steps, the model using SapBERT achieved an FID_{XRV} of 6.0 (RadBERT: FID_{XRV} of 6.7), compared with the random U-Net only model (FID_{XRV} of 16.5). A model fine-tuned on all available views (that is, projections from which an X-ray was taken, posterior–anterior (PA), anterior–posterior (AP) and lateral projections) yielded the lowest MS-SSIM among all the fine-tuned models.

Analysing the relationship between intra-prompt diversity and the length of the conditioning text prompt, it was found that fine-tuned models tend to show lower diversity as the text length increases. A considerably lower number of available training examples (at least for frontal CXR images) in the setting of longer prompts may contribute to this observation (Supplementary Fig. 1). This relationship was not observed for the model trained for only 1k steps, which could be explained by an inherent limitation of MS-SSIM as a metric for diversity.

Multi-label classification of synthetic CXRs

To evaluate how well synthetic CXRs conveyed diagnostically relevant information for deep learning models, a classification model trained on real CXRs (XRV[11]) was used to classify synthetic CXRs (Table 2), using the class labels of the in-distribution test set (D_{Test}) as ground truth. Synthetic CXRs were generated by prompting the respective fine-tuned models with the corresponding impression sections (one image per prompt), under the assumption that the text prompt essentially represented the same information as the natural language processing-derived class labels.

Classification performance was random for the original SD baseline (area under the receiver operating characteristic curve (AUROC) ~0.5) and poor for the few-shot DreamBooth baseline (AUROC 0.61). Fine-tuning for 1k steps already led to markedly higher performance

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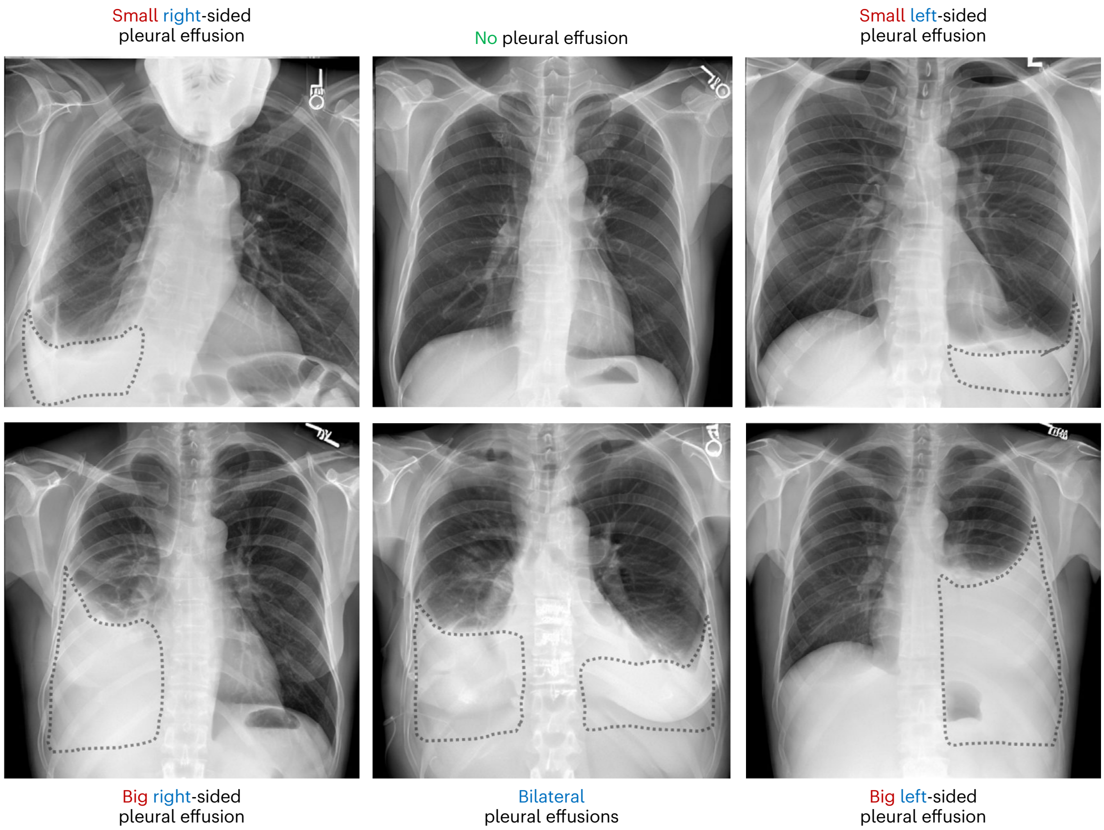

Fig. 3 | Text-conditional appearance of radiological findings. Here, presence or absence of a finding (pleural effusions, dotted regions of interest added for visualization) and dimensions such as size and laterality were controlled via prompting. Note that the model correctly incorporated the radiological convention of displaying the right patient side on the left side of the image, and vice versa. Each image was picked out of four generated CXRs per respective prompt. Colored text indicates modifiers for size (red), affected side (blue) and negation of a finding (green).

(average AUROC 0.83). Fine-tuning with a small learning rate (5×10^{-5}) for 1k steps achieved comparable average performance to training for 60k steps (average AUROC 0.82 versus 0.81).

Multi-modal factual correctness of synthetic CXRs

Classification performance, fidelity and diversity results were evaluated based on models that were solely trained on images, which may limit their ability to adequately capture the semantic correctness of the generated CXRs (for instance, if models exploited spurious image features). Thus, to analyse the outputs of different fine-tuned models at the vision–language intersection, we used established pre-trained multi-modal models that either generate text from CXRs or encode medical text. These models allow the evaluation of synthetic CXRs on the tasks of RRG, image–image and image–text retrieval.

The task of RRG consists of building systems that take CXRs as input and generate a text report describing image findings[14]. In this step, we used pre-trained RRG models to generate reports from synthetic CXRs and compared them with the underlying ground truth reports used to generate the same CXRs.

RRG

All fine-tuned SD models showed considerable improvements over the few-shot fine-tuned DreamBooth SD baseline for established domain-agnostic natural language generation metrics for semantic similarity with respect to content (for example, Bi-Lingual Evaluation Understudy with four-grams (BLEU-4)[15] of up to 5.5 versus 2.4 for the baseline), sequence (for example, recall-oriented understudy for gisting evaluation based on the longest common subsequence (ROUGE-L)[16] of up to 20.3 versus 12.8 for the baseline) and context (Bidirectional Encoder Representations from Transformers score (BERTScore)[17] of up to 42.3 versus 36.6 for the baseline) (Table 3, left columns; higher values are better). The same was true when using radiology-specific metrics for precision and recall of radiological findings (F1CheXbert (F1cXb)[18] of up to 46.5 versus 36.6 for the baseline) and representation of radiological concepts (for example, RadGraph[19] scores of 15.0 (1k steps) and 18.9 (60k steps) versus 9.0 (baseline)) (Table 3, right columns). Training on multiple views for 60k steps led to slightly higher (+2% to +5%) RRG scores than models trained solely on PA views.

Zero-shot image–image retrieval

In clinical radiology, reviewing images of similar cases can frequently boost diagnostic confidence. In a similar vein, image–image retrieval quantifies similarity by comparing generated CXRs (query images) to real CXRs (candidate images) with analogous findings. By employing a multi-modally pre-trained image encoder (ConVIRT[20]), this task processes both query and candidate images, ranks candidates based on cosine similarity to the query, and determines precision by comparing abnormality labels. Similar to the RRG task, all fine-tuning approaches improved the image–image retrieval performance over the DreamBooth baseline (retrieval precision up to 41.8% versus 24.2% for the baseline; Table 4, left column, and Supplementary Table 1). The model trained on multiple views outperformed models that generated PA images only, both further fine-tuning the text encoder and the U-Net for 60k steps, on all precision scores by 7–13%. Higher retrieval performance was generally achieved when both U-Net and text encoders were fine-tuned, however, fine-tuned models using frozen domain-specific text encoders outperformed models using a frozen CLIP text encoder, suggesting that unfreezing and fine-tuning domain-specific text encoders may be a promising research direction.

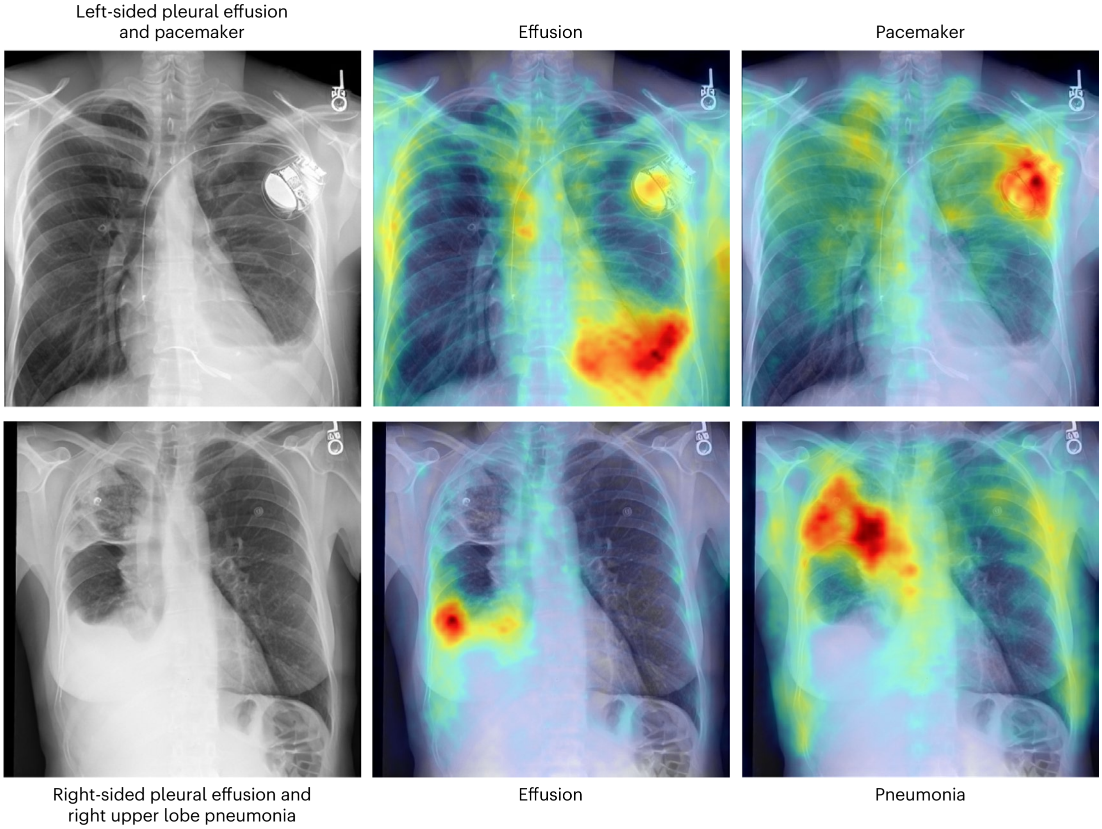

Fig. 4 | Combination of multiple abnormalities in synthetic CXRs. The left column contains images created by using the respective caption as prompt, while the middle and right columns highlight findings through an overlay of diffusion attentive attribution maps, which are created by upscaling and aggregating cross-attention word–pixel scores[60].

Zero-shot image–text retrieval

This task is similar to the image–image retrieval scenario, with the difference that it maps a query image embedding to a textual space to fetch the most likely corresponding radiology report impression for the given image. We selected the CXR-RePaiR model for this experiment[21]. Similar to the image–image retrieval task, all fine-tuning approaches improved the retrieval performance over the few-shot fine-tuned DreamBooth baseline (up to a RadGraph score of 9.2 versus 6.4 for the baseline). Retrieval performance was similar for the default (CLIP, precision up to 33.2%, versus baseline 19.9%) and the domain-specific text encoder SapBERT (precision 33.1%), and slightly worse for RadBERT (precision 27.7%). See Table 4 and Supplementary Table 2 for details.

Synthetic CXRs as training data

A possible use case for synthetic data is the training of deep learning models without having to share original data, potentially alleviating privacy concerns inherent to medical data. To investigate the value of using synthetic CXRs as training data, a CNN (Densenet-121) was trained on real CXRs, synthetic CXRs, or both ('mixed'), using 1%, 10% or 100% of the available training data (Extended Data Fig. 2 and Supplementary Table 3).

Increasing amounts of training data generally improved classification performance for all training set compositions on all evaluation datasets. Purely synthetic training data led to slight drops in performance compared with the baseline of real training data on all test sets. However, when mixed with real CXRs, similar or slightly improved classification performance was observed, especially with smaller amounts of training data (up to +2.4% AUROC compared with the baseline for in-distribution data; Supplementary Table 4).

A similar effect was observed on the out-of-distribution datasets D_{CheXpert} and D_{VinDr}, where mixed training data showed comparable or slightly improved performance in the small and mid-sized training set runs (performance gain of up to 3.2% over the baseline for 1% of the training data on the CheXpert data). The beneficial effects in low- and mid-sized training data settings were more pronounced when synthetic data exceeded real training data (Extended Data Fig. 3), showing improvements of up to +4.5% AUROC for in-distribution data and up to +7.3% AUROC for out-of-distribution data when adding four times the amount of synthetic data, compared to using only real CXRs (Supplementary Table 5). This performance increase diminished when the amount of real CXRs was the same as the generative model had been trained on (that is, 100% $D_{\mathrm{Dev,PA}}$). These results indicate that augmenting real data with synthetic data has the potential to improve classifier performance.

Reducing the size of the training dataset used to train the underlying domain-adapted SD model from 35k to 5k led to slightly reduced classification performance on the in-distribution test set (AUROC −4.8% to −7.6%) and comparable performance on out-of-distribution datasets (AUROC −3.1% to +3.6%; Supplementary Table 6).

SSL on synthetic CXRs

SSL is a method for deriving meaningful representations from unlabelled data by pre-training models on auxiliary objectives instead of directly on a target downstream task[22]. This is particularly relevant in medical imaging, where abundant unlabelled data are available, but labelling is cost-intensive. We assessed the merit of synthetic CXRs generated by RoentGen for SSL pre-training using a technique that inputs two differently augmented versions of a CXR image into Siamese network branches to learn invariant features[23].

To evaluate the representation quality, the SSL pre-trained models (DenseNet-121) underwent linear probing[24], with only a classification layer fine-tuned on 1%, 10% or 100% of the available training data while

Table 1 | Quantitative assessment of image fidelity and diversity

Experiment	Steps	Fine-tuned components		FID ↓		MS-SSIM ↓
		U-Net	Text encoder	XRV	IncepV3	
Baselines						
Original SD (no fine-tuning)				47.7	275.0	0.11±0.05
DreamBooth SD (few-shot fine-tuning)		✓		19.5	122.4	0.36±0.14
Learning rate						
SD U-Net + CLIP, high LR	1k	✓	✓	6.1	64.6	0.29±0.06
SD U-Net + CLIP, low LR	1k	✓	✓	6.0	65.0	0.30±0.06
SD U-Net + CLIP, high LR	12.5k	✓	✓	6.4	66.9	0.25±0.10
SD U-Net + CLIP, low LR	12.5k	✓	✓	8.2	67.4	0.22±0.10
SD U-Net + CLIP, high LR	60k	✓	✓	7.4	85.5	0.29±0.14
SD U-Net + CLIP, low LR	60k	✓	✓	**3.6**	**54.9**	0.40±0.17
Initialization						
Rnd. U-Net + CLIP	1k	✓	✓	9.0	233.8	0.20±0.08
Rnd. U-Net + CLIP	60k	✓	✓	4.9	75.4	0.37±0.13
Rnd. U-Net only	60k	✓		16.5	114.2	**0.14±0.06**
SD U-Net only	60k	✓		9.2	85.5	0.19±0.09
Text encoders						
SD U-Net + RadBERT	1k	✓		8.3	227.2	0.27±0.06
SD U-Net + RadBERT	12.5k	✓		4.6	68.5	0.24±0.05
SD U-Net + SapBERT	60k	✓		6.0	72.0	0.29±0.07
SD U-Net + RadBERT	60k	✓		6.7	88.3	0.30±0.07
Multiple views						
SD U-Net + CLIP	60k	✓	✓	19.3	114.0	0.14±0.08

FID for features extracted from a CXR-pre-trained DenseNet-121 (XRV) or ImageNet-pre-trained Inception V3. Smaller FID indicates higher fidelity to the original image distribution. Smaller MS-SSIM indicates higher intra-prompt image generation diversity. Training setups use a low learning rate (5×10^{-5}) unless otherwise specified. The best scores achieved through fine-tuning are in bold. MS-SSIM values are provided as mean±standard deviation. CLIP, Contrastive Language-Image Pre-training (text encoder component); LR, learning rate; Rnd., random initialization; SD, Stable Diffusion (initialized with original weights).

keeping other model weights static (Supplementary Tables 7 and 8). The performance of models pre-trained on synthetic data was found to be comparable to those pre-trained on real CXR, with minor performance declines when real CXRs were used for supervised training (up to −1.8% AUROC), but a slight performance boost was observed with synthetic or mixed data, especially for D_{VinDr} (up to +2.5% AUROC compared with SSL pre-training on MIMIC).

In another experiment, models were fine-tuned fully supervised on varying amounts of training data, yielding similar results for synthetic and real CXR pre-training across both in- and out-of-distribution datasets (−1.6% to +1.8% AUROC; Supplementary Tables 9 and 10).

The performance of models pre-trained and fine-tuned on real data (AUROC 0.612–0.750; Supplementary Table 9) was comparable to models without SSL pre-training and fine-tuned on a mix of real and synthetic data (AUROC 0.616–0.737; Supplementary Table 4).

Discussion

The LDM SD, pre-trained on billions of natural image–text pairs, can be domain-adapted to generate high-quality CXRs. As established in this work, by exploring various fine-tuning approaches and evaluation settings, the best-performing model—RoentGen—allows fine-grained control over the generated CXRs by using free-form medical text prompts. RoentGen is able to render a diverse array of radiological findings (such as pleural effusions and pneumothoraces) beyond the restrictive class-conditional settings of prior generative methods.

Historically, GANs dominated medical image synthesis owing to their ability to generate synthetic images with notable realism but are notorious for their instability during training. A key advantage of diffusion models lies in their capacity for relatively stable training and producing image quality that is comparable or even superior to that of GANs. In a previous study, an LDM (Medfusion) achieved markedly higher image quality compared with state-of-the-art GAN implementations (FID: 84.31 for ProGAN and 28.69 for StyleGAN-3 versus 17.28 for Medfusion)[25]. Our exploration into the integration of medical terminology through text prompts for controlling the appearance of specific findings further illustrates the flexibility and potential of LDMs over traditional GAN-based approaches, which mostly require specific (re-)training in conditional generation settings. With RoentGen, the appearance of these findings can be modified using natural language (for example, by prompting for variations in size) including medical terminology (with findings like ‘consolidation’ or ‘pleural effusion’), leading to radiologically plausible synthetic CXRs. This can be an effective tool to create tailored synthetic data in scenarios of class imbalance, mitigating sources of bias and increasing fairness[26].

Pre-trained CNN-based classification performance was similar for synthetic CXRs from models fine-tuned for few (1k) and for many (60k) training steps. However, manual review and image fidelity metrics suggested that longer training generally increased image quality. A possible explanation is that early training refines global features necessary for classification, while longer training may refine local features that are more subtle but contribute to the overall appearance of the generated CXRs for human perception. Nonetheless, the employed metrics underscore the effectiveness of the proposed domain adaptation approach by capturing differences between out-of-domain

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Table 2 | Multi-label classification performance of a pre-trained CNN on D_{Test} and on 5k synthetic CXRs generated using the models listed in the experiment column

Experiment	Steps	Atelectasis	Cardiomegaly	Oedema	Pleural effusion	Pneumothorax	Macro-average
Baselines							
Real CXR (D_{Test})		0.75	0.84	0.84	0.87	0.78	**0.82**
Original SD		0.48	0.49	0.49	0.48	0.55	0.50
DreamBooth SD		0.62	0.54	0.50	0.72	0.66	0.61
Learning rate							
SD U-Net + CLIP, high LR	1k	0.76	0.82	0.85	0.90	0.84	**0.83**
SD U-Net + CLIP, low LR	1k	0.75	0.79	0.84	0.91	0.80	0.82
SD U-Net + CLIP, high LR	60k	0.72	0.82	0.69	0.86	0.73	0.76
SD U-Net + CLIP, low LR	60k	0.78	0.84	0.78	0.89	0.76	0.81
Initialization							
Rnd. U-Net + CLIP	1k	0.72	0.69	0.70	0.82	0.76	0.74
Rnd. U-Net + CLIP	60k	0.79	0.82	0.78	0.90	0.79	**0.82**
Rnd. U-Net only	60k	0.72	0.76	0.72	0.87	0.76	0.77
SD U-Net only	60k	0.77	0.82	0.77	0.90	0.74	0.80
Text encoder							
SD U-Net + RadBERT	1k	0.67	0.59	0.65	0.70	0.78	0.68
SD U-Net + RadBERT	12.5k	0.77	0.78	0.80	0.87	0.81	0.81
SD U-Net + RadBERT	60k	0.72	0.73	0.67	0.86	0.78	0.75
SD U-Net + SapBERT	60k	0.80	0.84	0.77	0.90	0.77	**0.82**
Multiple views							
SD U-Net + CLIP	60k	0.72	0.73	0.67	0.86	0.78	0.75

Corresponding impression sections in D_{Test} were used to generate the sample images. Only labels with an AUROC ≥0.75 on D_{Test} are shown. Training setups used a low learning rate (5×10^{-5}) unless otherwise specified. The best average scores for each experimental category are in bold. LR, learning rate; Rnd., random initialization; SD, Stable Diffusion (initialized with original weights).

Table 3 | Factual correctness metrics for the task of RRG

Experiment	Domain-agnostic scores			Radiology-specific scores	
	BL4↑	ROUGE-L↑	BERTScore↑	F1cXb↑	RadGraph↑
Real CXRs (D_{Test})	8.9	23.0	45.6	50.4	22.5
DreamBooth SD	2.4	12.8	36.6	36.6	9.0
SD U-Net + CLIP, 1k	3.6	15.8	38.0	46.2	15.0
SD U-Net + CLIP, 60k	**5.5**	**19.9**	**42.3**	**46.5**	18.9
SD U-Net + RadBERT, 60k	4.7	18.0	40.3	42.6	16.3
SD U-Net + SapBERT, 60k	4.9	**20.3**	**42.3**	46.2	**19.0**
SD U-Net only, 60k	4.1	15.6	38.7	36.3	13.5
Multiple views, 60k	7.0	22.5	43.2	46.3	20.1

The best scores achieved through fine-tuning are in bold. Training setups use a low learning rate (5×10^{-5}) unless otherwise specified. BL4, BLEU-4; SD, Stable Diffusion (initialized with original weights).

pre-trained baseline models, few-shot fine-tuned models and more extensively fine-tuned models. Overall, the best performance was achieved through jointly fine-tuning both the pre-trained SD U-Net and the CLIP text encoder. Replacing the general-domain CLIP text encoder with a frozen domain-specific text encoder improved the performance when training the U-Net from scratch, indicating a potential for training speed-up when aiming for domain adaptation.

Our findings support the potential of synthetic data in augmenting real training data. In the context of supervised learning, moderate gains on both in- and out-of-distribution evaluations were seen when real training data were augmented with the same amount of synthetic CXRs, up at +3.2% AUROC over the baseline of exclusively real training data. This beneficial effect was even more pronounced in low- and mid-range training data scenarios when adding a surplus of synthetic data, while a plateauing of the effect was observed when the amount of real training data matched the amount of data the generative model had been trained on. Slight performance declines were also observed when training classification models on purely synthetic data. Such behaviour suggests that, for supervised training, synthetic data can act as a dataset augmenter, rather than a complete replacement for real data. Future studies can explore if the extended compositionality provided by language guidance can push classification performance beyond the limits of using all available real training data.

Beyond the use as data augmentation resource, we evaluated the potential of synthetic CXRs in the context of SSL, which involves learning meaningful representations from data through pre-training on auxiliary tasks. SSL is a viable strategy in the realm of medical imaging, given the vast amounts of unlabelled medical data whose potential often remains untapped due to the high time and costs associated with annotation by domain experts[22]. To explore the benefits of SSL, we performed several experiments using real, synthetic and mixed datasets with a recent SSL pre-training strategy that has been optimized for CXRs[23]. Our findings suggest that augmenting real CXRs with synthetic CXRs can enhance classification performance at a level

Table 4 | Factual correctness metrics for the tasks of image–image and image–text retrieval

Experiment	Image–image retrieval	Image–text retrieval			
	Precision ↑	Precision ↑	BERTScore ↑	F1cXb ↑	RadGraph ↑
Real CXRs (D_{Test})	47.7	39.9	35.8	49.1	11.3
DreamBooth SD	24.2	19.9	31.8	35.0	6.4
SD U-Net + CLIP, 1k	34.0	28.2	33.8	42.7	6.8
SD U-Net + CLIP, 60k	**41.8**	**33.2**	**34.3**	45.3	**9.2**
SD U-Net + RadBERT	36.8	27.7	33.2	41.6	7.9
SD U-Net + SapBERT	37.8	33.1	34.0	**45.4**	8.2
SD U-Net only	34.7	26.2	32.6	39.0	8.0
Multiple views	45.6	38.3	34.4	47.4	10.2

The best score for each metric and each task setting is in bold. Precision indicates the retrieval precision (in %) for the k top results macro-averaged for $k \in \{5, 10, 50\}$. SD, Stable Diffusion (initialized with original weights).

comparable to SSL pre-training on an equivalent amount of real data. Perhaps more importantly, synthetic data can also be beneficial for SSL pre-training itself, demonstrating nearly comparable performance compared with pre-training on real data. This indicates that the feature representations learned on synthetic data during SSL pre-training rival those from real data. Remarkably, models SSL pre-trained on synthetic data and also fine-tuned solely on synthetic data showed considerable performance improvement across all datasets when scaling the size of the synthetic datasets.

The benefits of using synthetic data are particularly relevant because scaling with larger pre-training datasets is beneficial both in natural and medical imaging domains[27,28]. While real datasets have a fixed limit in terms of their scale, synthetic datasets provide an opportunity to scale the size of pre-training datasets, especially because they are capable of producing diverse inputs for similar prompts. Overall, these promising results support the idea of using synthetic data for multi-institutional machine learning without the need to share real data[29]. This could be an effective strategy to counter the paucity of well-curated medical imaging datasets and serve as a complementary or alternative solution to approaches like federated learning.

A medical image should depict anatomical and pathological structures as clearly and as truthful as possible, limiting the immediate use of synthetic images in the diagnostic process. However, a model that has learned the underlying data distribution through domain adaptation, making it capable of generating realistic CXRs, could perform tasks with more direct medical relevance. For instance, classic radiographic phenomena such as the (loss of) silhouette sign or specific pathologies could be illustrated and highlighted without consulting image databases (Extended Data Fig. 4). Other notable examples from the recent literature suggest the capability of pre-trained denoising diffusion models in zero-shot classification, out-of-distribution detection and segmentation tasks[30–32], indicating a broader scope of applicability for medical tasks. Building widely capable models on domain-specific, multi-modal datasets is one of the primary factors currently driving many research fields, including medicine[33,34]. We contribute by presenting a way to domain-adapt a large pre-trained model that can be further optimized across several specific use-cases. In fact, the release of RoentGen has already led to promising research results, such as the evaluation of large-scale vision–language pre-training[35] and creating counterfactual explanation methods[36].

Rigorous evaluation is an important cornerstone on the way of translating technical developments into practice. In addition to human domain expert (radiologist) evaluation, the proposed evaluation framework for domain-adapting vision–language models helps to assess the quality and factual correctness of the synthetic images by testing different tasks like RRG, image–image and image–text retrieval, allowing to compare different generative models on multiple levels.

Recently, diffusion models have been shown to be able to reproduce training data in some cases[37]. Due to the sensitive nature of medical data and to protect patient privacy, future studies will have to validate these findings and further evaluate the safety of such an approach[38]. Another future research direction is to investigate different fine-tuning strategies that allow to limit catastrophic forgetting (see also Supplementary Results, Supplementary Discussion and Supplementary Fig. 2) and to increase the potential benefit of leveraging knowledge previously acquired through extensive pre-training.

Although the presented approach proved effective for domain adaptation, some limitations remain. First, a single-institution dataset (MIMIC-CXR) was used to fine-tune RoentGen, which can be a source of bias. Measuring the cross-institutional performance and robustness of the model is necessary, and further training it on multi-institutional data could arguably increase its generalizability, improve its flexibility to conditioning prompts and increase the diversity of the synthetic images. To address this, more sufficiently large datasets of medical images and corresponding text reports are needed.

Second, the token length limit of the currently employed CLIP text encoder restricted training to the impression sections of the radiology reports associated with each image. These sections might not capture every aspect of the image, and enriching the textual information with other sections (like the findings section, or clinical details) could enable more fine-grained conditioning capabilities.

Third, radiologists noted that, while the text prompts were on average reflected in the image, especially longer prompts mentioning multiple findings were not always faithfully translated. Similar issues are observed in many current text-to-image models[39]. Inspired by successes in aligning large language models with human preferences, proposed solutions to increase text–image alignment include further supervised fine-tuning and the use of reward models trained on human feedback with[39] or without reinforcement learning[40]. Beyond improved text–image alignment, such fine-tuning techniques grounded in human feedback could enhance the quality of the generated images[41] (for example, in our context, favouring radiologically plausible samples). Additionally, the resulting reward models could also be used as (domain-specific) automatic image scoring functions[41]. Thus, learning from human feedback seems a promising avenue to address this current limitation of domain-adapted text-to-image models like RoentGen.

Fourth, we noticed that the model was prone to overfitting when trained on small datasets of a few hundreds of images. Follow-up studies that evaluate fine-tuning under limited data constraints could benefit from detailed measures of overfitting.

Outlook

Medical-domain adaptation of latent diffusion models for text-conditioned image synthesis is feasible and has considerable potential in medical imaging, where most image studies are paired with a text report. Text-conditioned synthetic CXRs can augment available training data without the need for additional resource-intensive manual labelling by domain experts.

Leveraging the extensive general domain pre-training that went into the creation of models like SD, our approach can be used to create domain-adapted models on specific hospital datasets with moderate resources. The method is probably transferrable to other 2D medical imaging modalities, provided that enough paired image and text data are available. It is also conceivable that the vision–language fine-tuning approach can be extended to three-dimensional modalities such as computed tomography and magnetic resonance imaging.

Methods

Datasets

In this study, three publicly available CXR datasets were used. We created two development sets ($D_{\mathrm{Dev,Multi}}$ and $D_{\mathrm{Dev,PA}}$), an in-distribution test set (D_{Test}) derived from the MIMIC-CXR dataset[10] (version 2.0.0) and two out-of-distribution test sets D_{CheXpert} and D_{VinDr} derived from the CheXpert[6] (version 1.0) and VinDr-CXR[42] (version 1.0.0) datasets. See Supplementary Table 11 for detailed summary statistics.

MIMIC-CXR consists of 377k CXRs and corresponding text reports obtained from 228k patients at the Beth Israel Deaconess Medical Center (Boston, MA, United States) under institutional review board approval. We used the impression section, a summary section of a radiology report focused on the relevant findings and their interpretation, excluding samples with length <7 characters and those exceeding the CLIP tokenizer's maximum length of 77 tokens (14% of all cases). MIMIC-CXR is organized into ten folders (p10–p19). We used CXR-report pairs from folders p10–p18 as development sets ($D_{\mathrm{Dev,Multi}}$ with multiple views (PA, AP and lateral view), $n = 176$k, and $D_{\mathrm{Dev,PA}}$ with only PA views from unique patients, $n = 38$k), and randomly sampled CXR-report pairs from folder p19 to create the test set (D_{Test}, $n = 5$k). Samples used in the official MIMIC test set were excluded from the training sets.

MIMIC-CXR includes class labels for 14 different radiological findings (for example, 'pneumothorax', including a 'No Finding' class), originally derived from the text reports[10]. Each class is assigned the label 1 if the findings is present, 0 if the finding is absent and −1 if the presence is uncertain. We treated the latter case as positive and empty fields as negative. To address class imbalance, we unde rsampled the majority class ('no finding') to approximately match the prevalence of the next most frequent abnormality class ('lung opacity', 24%).

To assess the generalizability of classification models trained on synthetic data to out-of-distribution datasets, we randomly sampled 5k PA-view CXRs from the CheXpert dataset (total $n = 224$k from 65k patients[6]) and 4k frontal-view CXRs from the VinDr-CXR dataset (total $n = 18$k, collected from two hospitals[42]). We refer to these additional test sets as D_{CheXpert} and D_{VinDr}, respectively. To mitigate label distribution differences across datasets, some classes were undersampled where possible (Supplementary Table 11).

Labels for the CheXpert dataset were derived with a natural language processing labelling technique assessing 14 radiological findings, the same set of labels that are also provided for MIMIC-CXR[6,10]. The VinDr dataset provides a slightly different set of labels, derived from manual annotations created by three to five radiologist readers[42]. At the time of the experiments, no text reports were publicly available for these datasets.

Fine-tuning of a pre-trained LDM

All fine-tuning experiments were conducted on a high-performance computing cluster using 64 NVIDIA A100 graphics processing units (GPUs). SD components were fine-tuned using CXRs at a resolution of 512 × 512 pixels, with a batch size of 256. Bfloat16 floating point precision was used. In this setting, fine-tuning a model took approximately 20 min for 1k training steps, 5 h for 12.5k training steps and 1 day for 60k training steps. Model weights for the SD pipeline (version 1.4) were obtained from the repository CompVis/stable-diffusion-v1-4 (Hugging Face hub). The implementation is based on the diffusers library[43] (version 0.7.1) and the ViLMedic library[44] (version 1.2.8).

Model architecture

This project used the LDM SD[1] (version 1.4) as the underlying model. SD is a pipeline with three main components (Fig. 1): a Kullback–Leibler-regularized variational autoencoder (VAE) trained on a combination of perceptual loss and a patch-based adversarial objective, capable of mapping high-dimensional (pixel space) image inputs into lower-dimensional latent representations; a denoising U-Net backbone augmented with a cross-attention component[45] interacting with the intermediate layers of the U-Net; and a conditioning mechanism in the form of a CLIP ViT-L/14 text encoder mapping text inputs to a 768-dimensional embedding space, with a limit of 77 tokens (CLIP tokenizer limit)[9].

In the following, unless otherwise specified, this architecture was not modified except for disabling a built-in 'safety checker', as it displayed a high false-positive rate for medical prompts.

To generate an image, a Gaussian noise sample initializes a random vector, which is iteratively denoised with SD's U-Net, augmented by a cross-attention mechanism that communicates with SD's pre-trained CLIP text encoder. The obtained denoised latent vector is mapped to pixel space by the VAE's decoder component.

For CXR synthesis, a classifier-free guidance scale of 4.0 and 75 inference steps with the default noise scheduler (pseudo numerical methods for diffusion models, PNDM) were used[46].

Baseline (few-shot fine-tuning)

Previous research proposed fine-tuning of the U-Net on target domain examples while keeping the weights of the text encoder and VAE frozen ('DreamBooth' approach)[47]. While this approach allows generating synthetic CXRs with simple pathologies controlled by text conditioning, it is prone to overfitting, and the generative diversity (assessed by visual comparison) is low[48]. In this work, the few-shot tuned SD model proposed in ref. 48, trained on 50 CXR–text pairs for a respective pathology, was used as a baseline (referred to as 'DreamBooth SD').

Fine-tuning experiments

For fine-tuning, four experimental dimensions were explored: (1) the effect of learning rate and number of training steps; (2) which combination of SD components is optimally fine-tuned or retrained; (3) switching the CLIP text encoder with different domain-specific text encoders; (4) the effect of training on all available views (PA, AP and lateral) versus training exclusively on PA views (default approach). Specifically, a high and a low learning rate (1×10^{-4}, 5×10^{-5}) were tested for a low, medium or high number of training steps (1k, 12.5k, 60k). The listed values were selected for practicality. Unfrozen model components were trained continuously.

In all of the following experiments, the VAE component weights were kept frozen, as previous work indicated that the VAE component is suitable for CXR generation without additional fine-tuning[48]. We fine-tuned on CXR-report pairs starting from the original SD weights (default approach) or by training from scratch using random initialization. The text encoder was either trained jointly with the U-Net (default approach) or kept frozen.

Provided that the tokenizer limits (77 tokens) imposed by the CLIP text encoder of the original SD pipeline are preserved, it can be replaced by another text encoder. Here, two publicly available medical domain-specific text encoders were used: RadBERT[12] and SapBERT[13].

Fine-tuning on CXR-report pairs was achieved by predicting the noise component of noisy latent image representations, calculating the mean squared error between sampled and predicted noise, backpropagating and updating unfrozen SD pipeline components.

More formally, in a dataset of corresponding image and text pairs $D = \{(x_{\mathrm{text}}, y_{\mathrm{pixel}})_i\}_{i=1}^{n}$ for each pair i and timestep t, random Gaussian noise $N_{i,t}$ gets sampled in the latent space of dimensions (h, w):

$$N_{i,t} \sim N\left(0_{h\times w}, I_{(h\times w)^2}\right) \tag{1}$$

This Gaussian noise is added to the latent representation of y_{pixel} created by encoding the image with the VAE's encoder component E_{V}.

The text prompt x_{text} is encoded by a text encoder E_{T}. Independent Gaussian noise $N_{i,t}$ is added to the latent representation of an image for a random number of timesteps T. The U-Net ψ processes the noisy latent representation $E_{\text{V}}(y_{\text{pixel}}) \bigoplus_T N_{i,t}$ and encoded prompt $E_{\text{T}}(x_{\text{text}})$ to predict the last added noise $\hat{N}_{i,T}$, where $\bigoplus_T$ refers to an iterative adjusted addition for T timesteps.

$$\hat{N}_{i,T} = \psi(E_{\text{T}}(x_{\text{text}}), E_{\text{V}}(y_{\text{pixel}}) \bigoplus\nolimits_T N_{i,t}, T) \tag{2}$$

A mean squared error loss computed between the true and predicted noises $N_{i,t}$ and $\hat{N}_{i,T}$ enables to compute gradients and update the weights of the unfrozen components:

$$\mathcal{L} = \frac{1}{h \times w} \sum_{k=0}^{h} \sum_{t=0}^{w} \left(\hat{N}_{i,T,k,l} - N_{i,T,k,l}\right)^2 \tag{3}$$

Radiological assessment

Two board-certified radiologists (M.P. and C.B., with 9 and 7 years of CXR reading experience) evaluated the quality of the generated CXRs at a resolution of 512 × 512 pixels through a web interface. In a first step, CXRs generated by prompting the best-performing model (RoentGen) with the literal CheXpert class labels (such as 'pleural effusion' or 'pneumonia') were analysed, accepting the representational quality if the finding was clearly visible and radiologically sound. Second, the class labels were augmented by adding modifying descriptors pertaining to the appropriate attributes of the findings, such as the laterality ('left', 'right', 'bilateral') and the extent of a finding ('big pleural effusion', 'small pleural effusion'). Third, it was tested if multiple radiological concepts could be combined (for example, 'big right-sided pleural effusion and right-sided upper lobe pneumonia').

Finally, the two radiologists read 107 pairs of synthetic images and corresponding original impression sections (representatively sampled considering the CheXpert class label distribution), rating the results on a 5-point Likert scale (−2: the synthetic image does not correspond at all to the conditioning prompt; −1: the image does not show all of the findings mentioned in the impression section; 0: the image shows the most salient finding of the prompt but not all aspects; 1: the image shows most of the aspects mentioned in the prompt, 2: the synthetic image aligns perfectly with the prompt). The scores were averaged and reported as mean ± standard deviation.

Generative fidelity and diversity

The FID is a metric that assesses the distance between the feature vectors for real and generated images and is usually calculated using the 2,048-dimensional activation vector from the pool3 layer of an Inception V3 model[49] pre-trained on ImageNet, which might not be appropriate in a domain adaptation setting[50]. For this reason, FID scores were additionally calculated from intermediate layers of an in-domain classification model trained to detect common pathologies in CXRs (DenseNet-121, XRV, 1,024-dimensional feature vector)[11].

Generative diversity was assessed by calculating the pairwise MS-SSIM (Gaussian kernel size of 11, sigma of 1.5 (ref. 51)) of ten synthetic CXRs per impression section of D_{Test} and averaging the results. A lower MS-SSIM indicates a smaller structural similarity between images and can be interpreted as higher diversity among the outputs[52].

Multi-label classification

All real CXRs from D_{Test} and the synthetic CXRs generated from the D_{Test} impression sections were classified using a pre-trained CNN model (DenseNet-121, XRV) capable of detecting a range of typical CXR abnormalities[11]. Following the model's original specifications, all (real and synthetic) CXRs were pre-processed by centre-cropping and downsizing (224 × 224 pixels). Results are reported as AUROC.

RRG

To evaluate the factual correctness of generated images, an RRG model pre-trained on MIMIC-CXR was used[19]. The evaluation was carried out as follows: (1) synthetic CXRs were generated by fine-tuned SD models for every ground-truth impression section of D_{Test}; (2) these generated CXRs were used as input to the pre-trained RRG model that outputs new impressions; (3) these new impressions were then compared with the ground-truth impression used to generate the images. The RRG model's performance was also reported when using the ground-truth images as an upper-bound baseline.

To evaluate the newly generated impression against the ground-truth impressions, established natural language generation metrics (BLEU-4 (BL4)[15], ROUGE-L[16] and BERTScore[17]) and RadGraph score[19], a factual-oriented metric, were used. The latter metric evaluates the correctness, completeness and consistency of the generated impression using named entity recognition and entity relation systems trained on radiologist-annotated data. Additionally, the F1cXb[18] metric computing the F1 score between the prediction of CheXbert[53] (a model trained for the extraction of 14 common observation labels from text reports) run on the generated report and the ground truth report is reported (3). For further details on the metrics, see Supplementary Methods.

Zero-shot image–image retrieval

As pre-trained visual encoder, ConVIRT, a multi-modal model trained using contrastive learning methods using radiology reports and CXRs, was used[20]. A total of 200 query images and 400 candidate images were randomly selected from a subset of CXRs acquired by filtering the MIMIC-CXR test set to images featuring a single abnormality (atelectasis, consolidation, cardiomegaly, oedema, fracture, pleural effusion, pneumonia or pneumothorax). Synthetic CXRs generated by different models using MIMIC-CXR impression sections as prompts served as query images, real CXRs corresponding to the reports as candidate images. Retrieval performance was assessed in the form of average retrieval precision (Precision@k, where $k \in \{5, 10, 50\}$).

Zero-shot image–text retrieval

For this experiment, we chose the CXR-RePaiR model[21], a retrieval-based RRG approach using a pre-trained contrastive language-image model. This model was trained with an approach similar to CLIP, maximizing the similarity between the ground-truth text and image pairs embeddings. Retrieval performance was assessed using the Precision@k metric, indicating how many matching results are within the top k results. A point of precision was given if a retrieved candidate impression had the same class label as the query image. BERTScore[17], RadGraph score[19] and F1cXb[18] can be used to compare the top-1 retrieved impression against the corresponding ground-truth impression section (the one used to generate the synthetic CXR) (Table 4).

Synthetic CXRs as training data

For this experiment, an ImageNet-pre-trained DenseNet-121 was trained on 1%, 10% or 100% of 35,000 CXRs drawn from $D_{\text{Dev,PA}}$ that consisted of only real CXRs, only synthetic CXRs or an equal amount of real and synthetic data ('mixed'). To investigate the effect of scaling the amount of synthetic data beyond the available training data, CXRs obtained by prompting RoentGen multiple times (up to 4) with the same impression sections were used as additional training data. The SD model used to generate the synthetic CXRs for most experiments in this section was fine-tuned on 35k PA CXRs for 60k training steps and prompted using impression sections from $D_{\text{Dev,PA}}$. Additionally, to analyse the effect of the SD fine-tuning dataset size, we trained a model on 5k samples for 10k training steps. The training sets for the respective sizes were created by building stratified splits with 1k samples each using iterative stratification[54,55].

All CXRs were centre-cropped and resized to a resolution of 224 × 224 pixels. No data augmentation technique was used. All classification models were trained using an AdamW optimizer (learning rate 0.0001, weight decay 0.05), for a maximum of 100 epochs. The model checkpoints with the highest validation AUROC were used to classify all samples from D_{Test}, D_{CheXpert} and D_{VinDr}. Every training and evaluation run was repeated five times with different initialization seeds. Results are reported as mean AUROC ± standard deviation and, for synthetic-only and mixed training runs, as percentage change compared with the respective baseline (real-CXRs-only training run of the same training set size).

Synthetic CXRs for SSL

To investigate the value of synthetic CXRs for SSL, we used a recently proposed SSL approach that leverages a Siamese network to create representations invariant to transformations[23].

SSL pre-training

An SSL method based on SimSiam[56] and previously tested for the use on CXRs[23] was chosen to serve as pre-training strategy. Briefly, the approach uses two identical and weight-sharing network branches that each take an augmented version of the same CXR as input. Both versions are processed by an identical encoder network (based on DenseNet-121). The resulting feature vectors are passed on to a two-layered multi-layer perceptron projector network that produces a latent representation z_i of the data. The latent representations produced by each branch are input to a predictor network h_i, a multi-layer perceptron that aims to predict the projection z_i of the opposing branch (that is, h_1 aims to predict z_2, while h_2 aims to predict z_1). The negative cosine similarity between the predictions of the predictor networks and the actual projected feature vectors is used as loss function.

The effectiveness of this approach depends on the set of augmentations employed in the network branches. As augmentation strategy, we used symmetrical dual-branch augmentations of random resized cropping (scale 0.3–0.9) followed by a random contrast and brightness adjustments (via the ColorJitter class of the torchvision library, with brightness and contrast factor arguments set to 0.5). This combination was previously[23] shown to perform comparably well on CXR data as other established SSL pre-training strategies such as SimCLR[57] and MoCo V2 (ref. 58).

Pre-training on $D_{\mathrm{Dev,PA}}$ was done using eight NVIDIA A100 GPUs on a single node with 32-bit floating point precision, using a stochastic gradient descent optimizer (learning rate 0.005, weight decay 0.0001, momentum 0.9) with a cosine decay learning rate scheduler and a batch size of 256 for 100 epochs.

Linear probing and supervised fine-tuning of SSL-pre-trained models

For linear probing, a single fully connected classification layer was trained on top of the otherwise frozen SSL-pre-trained models and evaluated on the in- and out-of-distribution sets. An AdamW optimizer (learning rate 0.0001, weight decay 0.05) with a cosine decay learning rate scheduler and a batch size of 256 was used. All experiments were done on eight NVIDIA A100 GPUs on a single node with 32-bit floating point precision. The performance is assessed as macro-averaged AUROC ± standard deviation for the labels shared by all evaluation datasets. The baseline for the percentage difference are models trained on real CXRs with an otherwise unchanged approach. Fully supervised training used the same parameters and setup as the linear probing experiments, with the exception that all layers of the SSL-pre-trained models were trainable during the procedure.

Statistical analysis

Results were processed in Python (version 3.10.4), using the libraries numpy (version 1.23.1), scipy (version 1.9.1) and pandas (version 1.4.4). Continuous variables are expressed as mean ± standard deviation and categorical variables as count or percentage. Classification performance is expressed as AUROC and calculated using the torchmetrics library (version 0.11.0). Classification performance for classifiers trained from scratch on real, synthetic or mixed data was averaged across five runs with different random number generator initialization seeds (Supplementary Table 3) and additionally macro-averaged across all applicable labels for Supplementary Tables 4–10.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The main data supporting the results in this study are available within the article and its Supplementary Information. The MIMIC-CXR dataset[10] (version 2.0.0) is available on PhysioNet after a credentialing process (https://doi.org/10.13026/C2JT1Q). The CheXpert dataset[6] (version 1.0) is available from https://stanfordmlgroup.github.io/competitions/chexpert/. The VinDr-CXR dataset[42] (version 1.0.0) is available on PhysioNet after a credentialing process (https://doi.org/10.13026/3akn-b287). Source data are provided with this paper.

Code availability

The model weights of RoentGen can be requested by researchers credentialed for MIMIC-CXR access. Instructions on how to obtain the model weights and the code to fine-tune Stable Diffusion 1.4 on a dataset of medical text–image pairs can be accessed via GitHub at https://github.com/StanfordMIMI/RoentGen. A model card[59] for RoentGen is available (Supplementary Table 11).

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Author contributions

C.B., P.C. and A.S.C. conceived and co-led the project and drafted the initial paper. P.C. and C.B. fine-tuned SD for CXR generation. M.P. and C.B. conducted the reader studies. C.B. designed, implemented and conducted the experiments pertaining to generative fidelity and diversity, classification, and supervised and self-supervised training on synthetic data. J.B.-D. designed, implemented and conducted the evaluation of multi-modal factual correctness. R.v.d.S., T.M.A. and S.P. provided additional technical advice and assistance. C.B., P.C., J.B.-D., R.v.d.S., M.P., J.M.Z.C., T.M.A., S.P., C.P.L. and A.S.C. helped revise the paper. A.S.C. supervised the project.

Competing interests

T.M.A. and S.P. are employees of Stability AI. C.P.L. reports activities not related to the present article: Board of directors and shareholder, Bunkerhill Health 3/31/2019, Option holder, Whiterabbit.ai 10/01/2017, Advisor and option holder, GalileoCDS 05/01/2019, Advisor and option holder, Sirona Medical 07/06/2020, Advisor and option holder, Adra 09/17/2020, Advisor and option holder, Kheiron 10/21/2021, Paid consultant, Sixth Street 02/07/2022, Paid consultant, Gilmartin Capital 07/18/2022. Recent grant and gift support paid to C.P.L.'s institution: BunkerHill Health, Carestream, CARPL, Clairity, GE Healthcare, Google Cloud, IBM, Kheiron, Lambda, Lunit, Microsoft, Nightingale Open Science, Philips, Siemens Healthineers, Stability.ai, Subtle Medical, VinBrain, Visiana, Whiterabbit.ai, Lowenstein Foundation, Gordon and Betty Moore Foundation. A.S.C. discloses consulting services to Patient Square Capital, Elucid Bioimaging, Skope MR, Culvert Engineering, Edge Analytics, Image Analysis Group and Chondrometrics GmbH; and is shareholder in LVIS Corp., Subtle Medical and Brain Key. The other authors declare no competing interests.

Additional information

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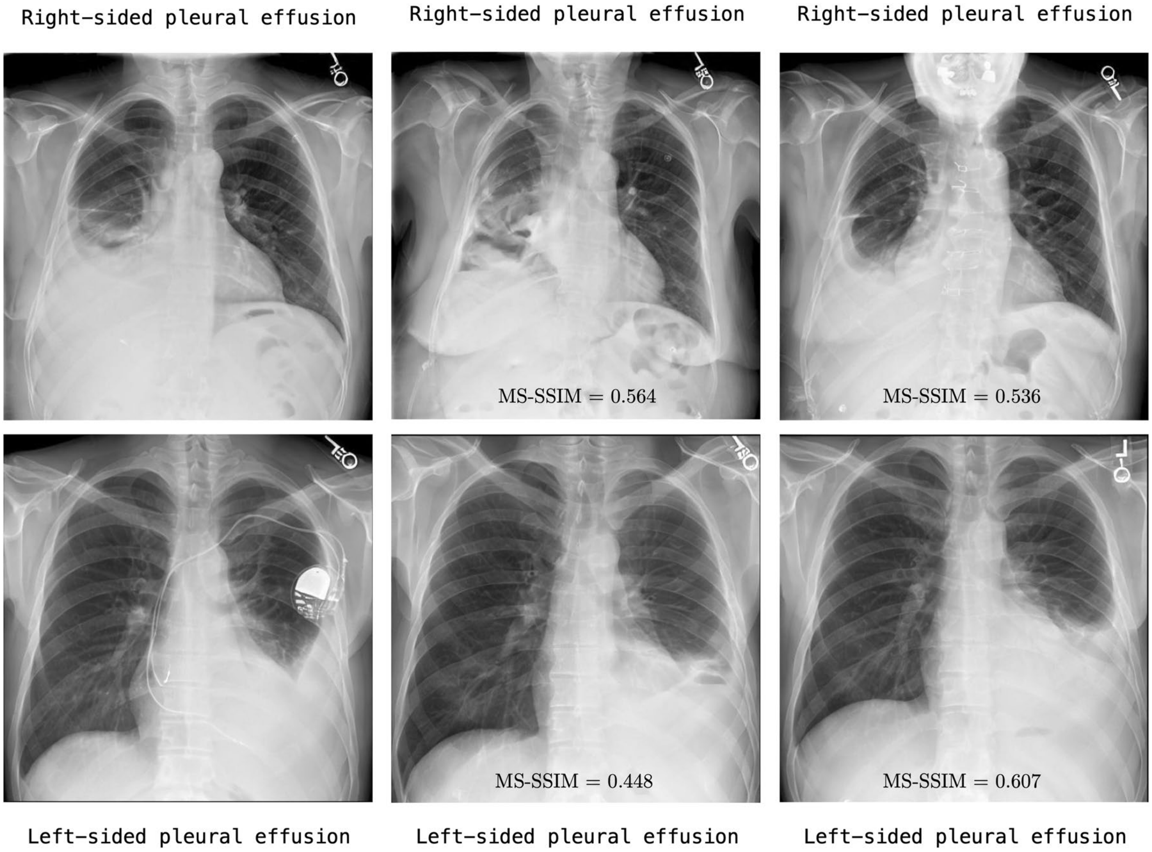

Extended Data Fig. 1 | Intra-prompt diversity of synthetic CXRs. Three generated samples per prompt are shown, using the prompts "Right-sided pleural effusion" (top row) and "Left-sided pleural effusion" (bottom row). Numbers on the three right columns represent pair-wise MS-SSIM values calculated between each image and the first image.

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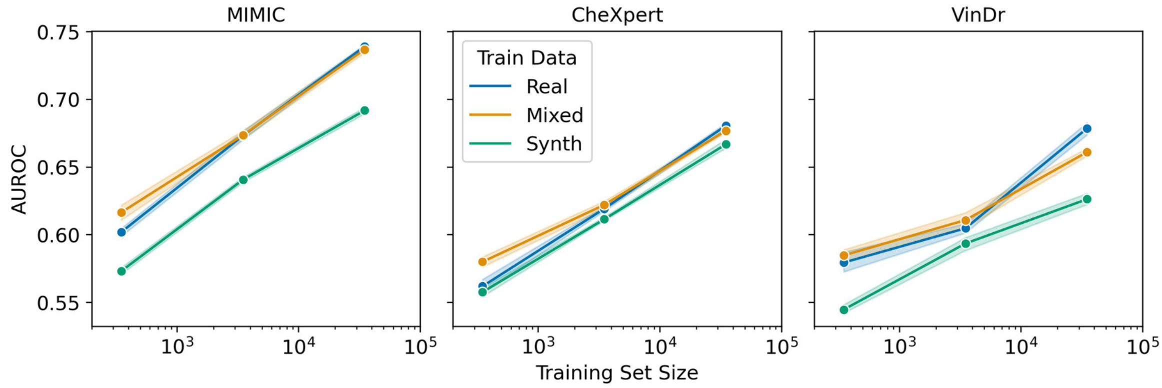
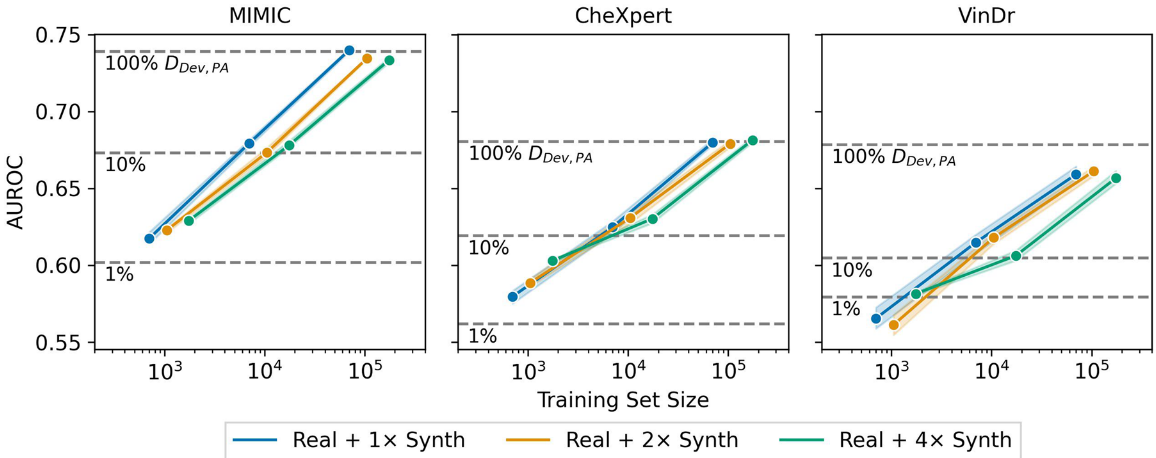


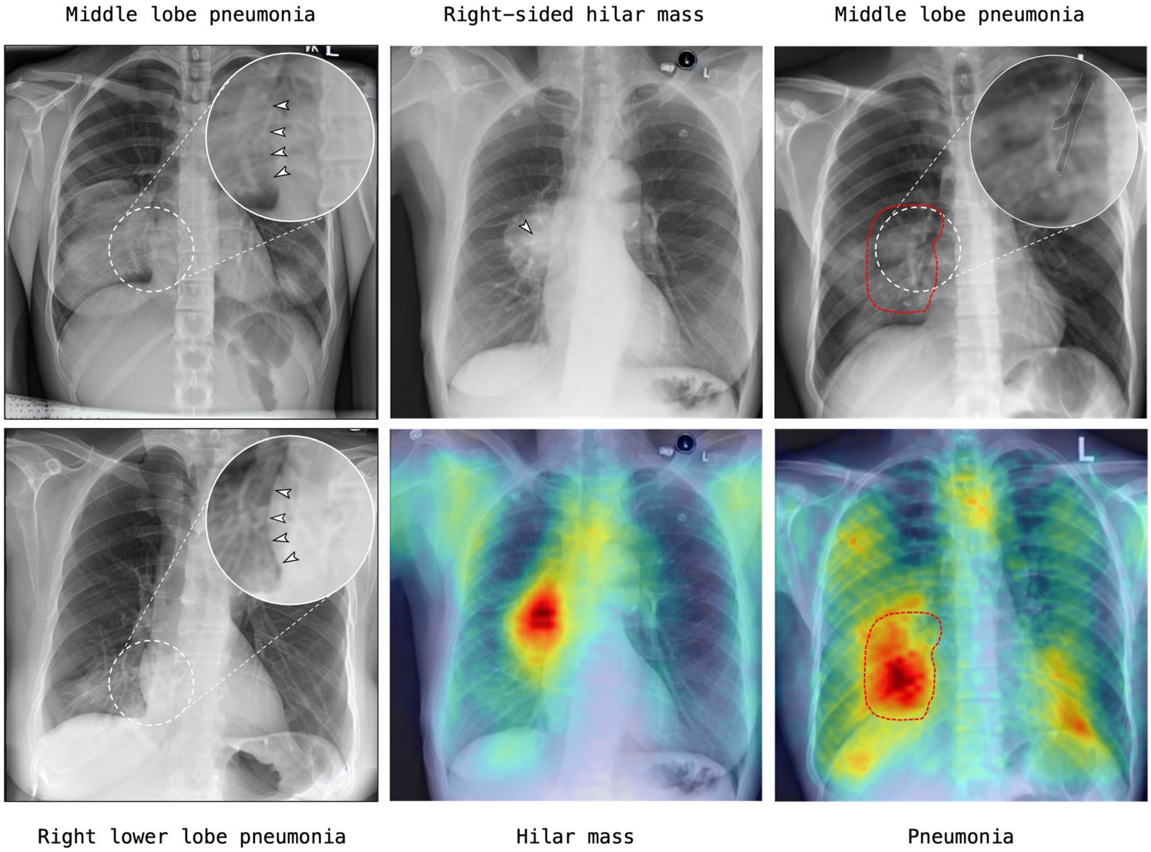
Extended Data Fig. 2 | Multi-label classification performance of DenseNet-121 trained on increasing amounts of real, synthetic or a blend of real and synthetic CXRs. AUROC is macro-averaged across the labels available for all test datasets (Atelectasis, Cardiomegaly, Consolidation, Pleural Effusion, Pneumothorax, Pneumonia, No Finding). Light areas indicate 95% confidence intervals.

 https://doi.org/10.1038/s41551-024-01246-y



Extended Data Fig. 3 | Effect of additional synthetic training data on multi-label classification performance. DenseNet-121 was trained on real and varying amounts of synthetic CXRs, which were obtained by sampling *RoentGen* multiple (2-4) times per prompt, using 1%, 10% and 100% of $D_{Dev,PA}$ as underlying dataset and text prompt source. The dashed horizontal lines indicate the classification performance of a model trained solely on real CXRs. AUROC is macro-averaged across the labels available for all test datasets (Atelectasis, Cardiomegaly, Consolidation, Pleural Effusion, Pneumothorax, Pneumonia, No Finding). Light areas indicate 95% confidence intervals.

 https://doi.org/10.1038/s41551-024-01246-y



Extended Data Fig. 4 | Educational use of RoentGen to illustrate signs and pathologies. Left column, Synthetic CXRs created with the prompts "middle lobe pneumonia" and "right lower lobe pneumonia", showcasing the "loss of silhouette sign" that can help to distinguish middle lobe pneumonia (silhouette absent) from lower lobe pneumonia (silhouette present) on CXR. **Center column**, Synthetic CXR showing an opacity at the right hilus, and corresponding diffusion attentive attribution map (DAAM) for the expression "hilar mass". **Right column**, Synthetic CXR showing another example of discrete middle lobe pneumonia with an air bronchogram (that is, visible endobronchial air against a background of increased lung opacity, top right image), with the main area of the DAAM highlighting the corresponding opacities in the middle lobe (bottom right image).

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Software and code

Policy information about availability of computer code

Data collection	No software was used to collect the data. The underlying data stemmed from publicly available datasets (MIMIC-CXR, CheXpert, VinDr-CXR; MIMIC-CXR and VinDr-CXR) are available upon completing a credentialing process.
Data analysis	The code used to domain-adapt Stable Diffusion is based on the diffusers library (version 0.7.1), and is available on GitHub at https://github.com/StanfordMIMI/RoentGen. Other evaluation and support tasks used in this work were implemented with custom Python code (version 3.10) using (among others) the Python libraries numpy (version 1.23.1), scipy (version 1.9.1), pandas(version 1.4.4), torchmetrics (0.11.0) and ViLMedic (version 1.2.8).

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The main data supporting the results in this study are available within the paper and its Supplementary Information. Source data for Extended Data figures are provided with this paper. The MIMIC-CXR dataset10 (version 2.0.0) is available on PhysioNet after a credentialing process (https://doi.org/10.13026/C2JT1Q). The CheXpert dataset6 (version 1.0) is available from https://stanfordmlgroup.github.io/competitions/chexpert/. The VinDr-CXR dataset43 (version 1.0.0) is available on PhysioNet after a credentialing process (https://doi.org/10.13026/3akn-b287).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	The datasets used in this study include publicly reported sex distributions.
Reporting on race, ethnicity, or other socially relevant groupings	Information on race, ethnicity or other socially relevant groupings was not available for all included datasets.
Population characteristics	MIMIC-CXR consists of 377k CXRs and corresponding text reports from 228k obtained from 65k patients at the Beth Israel Deaconess Medical Center Emergency Department (Boston, MA, USA) under institutional review board approval. Population characteristics are available after a credentialing process (https://www.nature.com/articles/s41597-019-0322-0). CheXpert is a public dataset for chest-radiograph interpretation, consisting of 224k chest radiographs of 65k patients who visited the inpatient and outpatient centers of the Stanford Hospital (Stanford, CA, USA) between October 2002 and July 2017. Population characteristics are available in the dataset documentation (https://arxiv.org/abs/2105.03020). VinDr-CXR is a dataset of 18k radiographs acquired in posterior–anterior projection, retrospectively collected from the Hospital 108 and the Hanoi Medical University Hospital (HMUH) in Vietnam. Population characteristics are available in the corresponding publication (https://www.nature.com/articles/s41597-022-01498-w/tables/3).
Recruitment	No participants were recruited for this retrospective study.
Ethics oversight	Data from the publicly available datasets was handled according to the requirements of the data-usage agreements. No additional ethical approval was needed for this retrospective study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

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Sample size	The development-set sample size was a result of filtering the MIMIC-CXR dataset to CXR taken in posterior–anterior projection and with a corresponding radiology-report impression-section length (after CLIP tokenization) within the limits of the tokenizer. The test set sample sizes of 5k for MIMIC-CXR and CheXpert and of 4k from VinDr-CXR were chosen for practicality.
Data exclusions	Data included in the official MIMIC-CXR test set were excluded from the development set. No other data exclusions were made.
Replication	The presented model is based on the publicly available Stable Diffusion model (version 1.4), whose weights can be obtained through the huggingface.com hub.
Randomization	Test datasets were randomly sampled from the available data. We undersampled the majority class ('No finding') to approximately match the next-most prevalent class.
Blinding	The investigators were not blinded during data collection, as it was not suitable for the chosen experimental design. For the reader study, only

Blinding	synthetic CXRs were being evaluated, and part of the evaluation was to determine whether radiological findings mentioned in the input prompt were represented in the resulting output.

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☒	☐ Flow cytometry
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